Interpreter

Season: 3 · Behavioral · Category: GoF · Level: Intermediate

Intent

- Map a grammar to classes that interpret sentences.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Spreadsheet formula AST.

When to Use

- See Intent.
- The design force named in Intent shows up more than once.

When Not to Use

- Overkill for one-off behavior.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.gof.behavioral.interpreter;

import java.util.Map;

public sealed interface Expression permits NumberExpr, AddExpr, MulExpr {
    int evaluate(Map<String, Integer> ctx);
}

record NumberExpr(int value) implements Expression {
    public int evaluate(Map<String, Integer> ctx) { return value; }
}

record AddExpr(Expression left, Expression right) implements Expression {
    public int evaluate(Map<String, Integer> ctx) {
        return left.evaluate(ctx) + right.evaluate(ctx);
    }
}

record MulExpr(Expression left, Expression right) implements Expression {
    public int evaluate(Map<String, Integer> ctx) {
        return left.evaluate(ctx) * right.evaluate(ctx);
    }
}
```

Modern Java 25 / Functional

- Java 25 pattern matching often replaces Visitor/Interpreter boilerplate.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: gof/behavioral/interpreter/
```

Pitfalls

- Prefer sealed + switch over classic Visitor in Java 25.
- Strategy as a function value.
- Observer: unsubscribe to avoid leaks.

Interview Q&A

- Q: Key interview angle for Interpreter? A: Name the force (undo, open/closed, open for extension) and a modern alternative.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Map a grammar to classes that interpret sentences.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Spreadsheet formula AST. [0:40] Intent — State the design force in one sentence: Map a grammar to cl...